

Professional Practices in IT
(CS-4001)

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Course Instructor(s)

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Final Exam

Total Time (Hrs.): 3

Total Marks: 90

Total Questions: 6

Roll No

Section

Student's Signature

Do not write below this line

Attempt all the questions.

Instructions

- This is an open-book exam. You are allowed to use your notes, reference material etc. in hard copy.
- No mobile phones, laptops, or other gadgets are allowed.

CLO #1: Understand the general context in which professionals work – the law and how it is created and the professions and professional bodies in computing. Evaluate the professional code of ethics from the ACM, the IEEE Computer Society, and other organizations.

[Time: 10 minutes]

Q.1: A software development company is facing two ethical dilemmas: [4+6=10 Marks]

Scenario A: The company's IT security team discovers that their employee monitoring software tracks bathroom breaks and personal conversations. Employees were never informed about the extent of this surveillance.

Scenario B: A junior developer with a disability requests specialized ergonomic equipment and screen-reading software to perform their job effectively. The company hesitates, citing budget constraints.

Using the concepts of negative and positive rights:

Answer each question maximum four lines; exceeding will incur negative marks.

- Identify which scenario primarily involves a negative right and which involves a positive right.
- Explain how the company should respond to each scenario to respect professional ethics and employee rights.

CLO #3: Analyze the project management techniques for IT projects to initiate, plan, execute and evaluate a project and Startups. understand and apply the essential numerical skills required for bookkeeping and accounting.

[Time: 45 minutes]

Q.2: Sara Traders provided the following information for the year ended 31 December 2024:

Additional Information

[3+2+5=10 Marks]

- Capital at start of year: Rs. 70,000

Income Data

- Sales Revenue: Rs. 200,000
- Cost of Goods Sold: Rs. 120,000

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- Operating Expenses: Rs. 45,000
- Assets & Liabilities (end of year)**
- Cash: Rs. 30,000
 - Accounts Receivable: Rs. 25,000
 - Inventory: Rs. 35,000
 - Equipment: Rs. 85,000
 - Accounts Payable: Rs. 40,000
 - Bank Loan: Rs. 30,000

Required:

- Prepare the **Income Statement** for the year.
- Calculate the **Closing Capital**.
- Prepare the **Balance Sheet** as at 31 December 2024.

Q.3: Calculate a machine costing Rs. 40,000/- over a period of 5 years on 10% Discount Factor. Annual maintenance is Rs.100/-. Calculate the NPV. **[5 Marks]**

Q.4: A boutique furniture workshop employs a team of craftspeople to produce two product lines: Standard and Premium dining tables. Each worker's total employment cost (salary + benefits + equipment allowance) and working schedule provides approximately 1,920 productive hours per worker per year. The company currently employs 3 workers. Management forecasts total sales of 380 dining tables this year and must recover annual fixed overheads of £72,000 from sales revenue. The two product lines vary in material costs, production time, and projected demand as follows:

- **Standard:** material cost = £120; production time = 8 hours. Expected sales = 240 units.
- **Premium:** material cost = £280; production time = 15 hours. Expected sales = 140 units.

Assume the direct labor hourly rate (wages + employer national insurance contributions) is £22.50 per hour. Show every step and intermediate calculation. **[5+5=10 Marks]**

- Calculate the overhead proportion to total cost direct.
- Calculate the overhead proportion to Labor content.

CLO #2: Explain the specific legal issues including software contracts and liability, intellectual property rights and legislation that affects the way in which computers are used or misused.

[Time: 75 minutes]

Q.5:

- Logitech Solutions signed a contract with City Bank to develop a loan processing system for PKR 500,000, to be delivered by September 2025. During development, LogiTech used proprietary code belonging to a competitor without permission. As a result, the competitor sued City Bank for PKR 200,000. Additionally, LogiTech delayed the project by three months, causing City Bank a loss of PKR 80,000 in operational efficiency. The dispute has continued for six months, incurring PKR 50,000 in legal costs and damaging the business relationship.

Explain how three specific contract clause: (i) Indemnity, (ii) Penalty Clauses, and (iii) Software

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Arbitration- could have protected both parties and resolved this situation more effectively. For each clause, state its purpose and how it would apply to this scenario.

Answer each clauses maximum 5 lines; exceeding will incur negative marks.

[10 Marks]

- ☒ b) A Digital forensic analyst investigating the illegal distribution of copyrighted movies over a peer-to-peer network. During the investigation, an individual who has downloaded multiple unauthorized copies of films and has also shared these files with other users online. Explain how UK legislation would apply to this situation. Discuss the **legal implications in detail**, and the potential consequences for the individual involved. **[10 Marks]**

- ☒ c) Spanish lawyers send a defamation claim to: "One Post, Three Countries." While on holiday in Turkey, a German student (logged through the hotel's Turkish ISP) tweets in English that a Spanish coach "fixes match". The tweet is seen in Madrid, retweeted 5,000 times. **[10 Marks]**

- Twitter Ireland (EU headquarters)
- The student's German home ISP
- Turkey's largest ISP

Answer in short bullets (≤ 15 words each):

- ☒ 1. Under EU law, which country is the easiest place for the coach to sue?
- ☒ 2. Name one international rule that lets a Spanish court demand user data from Ireland.
- ☒ 3. Cite one real 2015-23 case where online defamation crossed borders.
- ☒ 4. Give one ethical reason the German ISP should (or should not) hand over the student's name.
- ☒ 5. Suggest one technical step Twitter could still take to soften the harm.

- ☒ d) Sarah Mitchell is an investigative journalist who discovers that a pharmaceutical company, DataMed Ltd, has failed to disclose clinical trial results showing serious side effects of a widely used medicine. Sarah obtained this information by submitting a Freedom of Information request to an NHS Trust that conducted the trials.

DataMed Ltd claims that the clinical trial data contains personal medical information of patients, that releasing the data would damage their commercial interests, and that publication would violate patient privacy.

Sarah argues that the information can be anonymized, that the public interest in medication safety is more important than commercial concerns, and that she has a right to publish the information as a journalist.

Identify the legislation i.e. **Data Protection Law 1998 or FOIA 2000** and answer accordingly.

Answer in maximum 6 to 7 lines; exceeding will incur negative marks.

[2.5 x 4 = 10 Marks]

- ☒ 1. What types of personal and sensitive information are contained in clinical trial data, and why does DataMed Ltd argue that disclosure would breach patient privacy?

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2. How could the clinical trial data be disclosed to Sarah while still protecting the identities and privacy of trial participants?
3. What is the National Health Service (NHS) Trust's legal duty to provide information under the Freedom of Information Act 2000, and what exemptions might DataMed Ltd rely on to prevent disclosure?
4. How does the public interest in patient safety influence whether the NHS Trust should disclose the clinical trial data despite DataMed Ltd.'s objections?

CLO #4: Present strategies for gaining confidence in managing projects through simple project planning and working in a team to create a project plan which includes key tasks, critical path, dependencies and a realistic timeline. Understand why and to what extent managers need to be aware of general human resources issues.

[Time: 35 minutes]

Q.6. Kamyar is the president of a small game development company named Artworks. Kamyar's company is developing three new games. Kamyar finds that one game is complete and ready to sell. It is very violent and demeaning to women. It would probably sell a million copies.

Answer the following question not more than 3 to 4-line reasoning: **[2.5 x 6 = 15 Marks]**

Answer in maximum 5 lines; exceeding will incur negative marks.

1. Kamyar needs to decide what to do with the completed game, which is violent and demeaning to women but has strong commercial potential. List possible courses of action, and provide the pros and cons for each. Then, suggest what Kamyar is likely to do and justify the reasoning.
2. Artworks requires all applicants for the sound designer position to pass a test that involves heavy lifting of 20 kg audio equipment. Very few female candidates can perform this. Is this an example of direct or indirect discrimination? Justify your answer.
3. Artworks has advertised a job stating: "We are hiring a young man as a junior game tester because we prefer energetic men." Identify whether this is direct or indirect discrimination. Explain your reasoning.
4. Ali has been employed as a 2D mid-level developer at Artworks for the past six months. Kamyar has decided to focus exclusively on 3D game development and, as a result, dismissed Ali. What type of dismissal does this situation represent?
5. Artworks wants to hire a new 3D developer. Kamyar asks HR to design a selection process to ensure the candidate has strong 3D modeling skills. Suggest three effective selection methods HR could use to assess technical competence for a 3D developer role, and briefly explain why each method is appropriate.
6. Artworks wants to upgrade the skills of a current 2D developer to work on 3D modeling projects. What steps should HR take to ensure effective training and development for this employee? Include considerations for cost, skill assessment, and performance evaluation.

- Good luck -